

LÍMITES NOTABLES.

29)

$$\lim_{x \rightarrow 0} \frac{\sin ax}{ax} = 1$$

$$\lim_{x \rightarrow 0} \frac{\sin 2x}{x} = \quad \rightarrow a = 2$$

Multiplicamos numerador y denominador de la función por "2" en este caso:

$$= \lim_{x \rightarrow 0} \frac{2 \sin 2x}{2x} = \text{el límite de una constante por una variable es igual a la constante por el límite de la variable}$$

$$= 2 \lim_{x \rightarrow 0} \frac{\sin 2x}{2x} \rightarrow \text{límite notable igual a } 1 = 2 \cdot 1 = 2$$

30) $\lim_{x \rightarrow 0} \frac{\sin 3x}{2x} =$ siempre debemos multiplicar y dividir por el coeficiente del argumento de la "función seno"

$$= \lim_{x \rightarrow 0} \frac{3 \sin 3x}{3 \cdot 2x} = \lim_{x \rightarrow 0} \frac{3 \sin 3x}{2 \cdot 3x} = \frac{3}{2} \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} = \frac{3}{2} \cdot 1 = \frac{3}{2}$$

31) $\lim_{x \rightarrow 0} \frac{10x}{\sin 3x} = \frac{10}{3}$

32) $\lim_{x \rightarrow 0} \frac{\sin x}{5x} = \frac{1}{5}$

33) $\lim_{x \rightarrow 0} \frac{\sin 2x}{x} = 2$

$$34) \lim_{x \rightarrow 0} \frac{\sin^2 3x}{x^2} = \lim_{x \rightarrow 0} \left(\frac{\sin 3x}{x} \right)^2 \quad \text{El límite de una potencia es igual a la potencia del límite.}$$

$$= \left(\lim_{x \rightarrow 0} \frac{\sin 3x}{x} \right)^2 = \left(\lim_{x \rightarrow 0} \frac{3 \sin 3x}{3x} \right)^2 = \left(3 \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \right)^2 =$$

$$= (3 \cdot 1)^2 = 9$$

$$35) \lim_{x \rightarrow 0} \frac{\sin^2 5x}{15x^2} = \frac{5}{3}$$

$$36) \lim_{x \rightarrow 0} \frac{\sin^3 2x}{x^3} = 8$$

$$37) \lim_{x \rightarrow 0} \frac{\sin x}{x^2} = \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \frac{1}{x} = \text{El límite de un producto es igual al producto de los límites.}$$

$$38) \quad \boxed{\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1}$$

$$\lim_{x \rightarrow 0} \frac{\tan 3x}{7x} = \frac{3}{7}$$

El límite de un cociente es igual al cociente de los límites.

$$39) \lim_{x \rightarrow 0} \frac{\tan^2 2x}{x \cdot \sin 3x} = \lim_{x \rightarrow 0} \frac{\tan 2x \cdot \tan 2x}{x \cdot \sin 3x} =$$

$$= \lim_{x \rightarrow 0} \frac{\tan 2x}{x} \cdot \lim_{x \rightarrow 0} \frac{\tan 2x}{\sin 3x} = \lim_{x \rightarrow 0} \frac{2 \cdot \tan 2x}{2x} \cdot \left(\lim_{x \rightarrow 0} \frac{\tan 2x}{x} \cdot \lim_{x \rightarrow 0} \frac{x}{\sin 3x} \right) =$$

$$= 2 \cdot \lim_{x \rightarrow 0} \frac{\tan 2x}{2x} \cdot \frac{\lim_{x \rightarrow 0} \frac{\tan 2x}{x}}{\lim_{x \rightarrow 0} \frac{\sin 3x}{x}} = 2 \cdot 1 \cdot \frac{\lim_{x \rightarrow 0} \frac{2 \tan 2x}{2x}}{\lim_{x \rightarrow 0} \frac{3 \sin 3x}{3x}} =$$

$$= 2 \cdot \frac{2 \lim_{x \rightarrow 0} \frac{\tan 2x}{2x}}{3 \lim_{x \rightarrow 0} \frac{\sin 3x}{3x}} = 2 \cdot \frac{2}{3} = \frac{4}{3}$$

$$\begin{aligned}
 40) \lim_{x \rightarrow 0} \frac{\sin^2 4x}{\sin^2 2x} &= \lim_{x \rightarrow 0} \left(\frac{\sin 4x}{\sin 2x} \right)^2 = \left(\lim_{x \rightarrow 0} \frac{\sin 4x}{\sin 2x} \right)^2 \\
 &= \left(\lim_{x \rightarrow 0} \frac{\frac{\sin 4x}{x}}{\frac{\sin 2x}{x}} \right)^2 = \left(\frac{\lim_{x \rightarrow 0} \frac{\sin 4x}{x}}{\lim_{x \rightarrow 0} \frac{\sin 2x}{x}} \right)^2 = \left(\frac{\lim_{x \rightarrow 0} \frac{4 \cdot \sin 4x}{4x}}{\lim_{x \rightarrow 0} \frac{2 \cdot \sin 2x}{2x}} \right)^2 \\
 &= \left(\frac{4 \lim_{x \rightarrow 0} \frac{\sin 4x}{4x}}{2 \lim_{x \rightarrow 0} \frac{\sin 2x}{2x}} \right)^2 = \left(\frac{4 \cdot 1}{2 \cdot 1} \right)^2 = \frac{16}{4} = 4
 \end{aligned}$$

$$41) \boxed{\lim_{x \rightarrow 0} (1+ax)^{\frac{1}{ax}} = e} \text{ LIMITE NOTABLE}$$

$$\lim_{x \rightarrow 0} (1+3x)^{\frac{1}{x}} =$$

$$t=3x \rightarrow x=\frac{t}{3}$$

$$\lim_{t \rightarrow 0} \left(1 + 3 \cdot \frac{t}{3} \right)^{\frac{1}{\frac{t}{3}}} = \lim_{t \rightarrow 0} (1+t)^{\frac{3}{t}} =$$

$$= \lim_{t \rightarrow 0} \left[(1+t)^{\frac{1}{t}} \right]^3 = \left[\lim_{x \rightarrow 0} (1+t)^{\frac{1}{t}} \right]^3 = e^3$$

$\left\{ \begin{array}{l} \text{Tomamos una nueva} \\ \text{variable } t, \text{ y le llama-} \\ \text{mos } t \text{ a } ax. \\ ax = 3x \quad t = 3x \\ x = \frac{t}{3} \end{array} \right.$

$$42) \lim_{x \rightarrow 0} (1-5x)^{\frac{1}{x}} = e^{-5}$$

$$43) \lim_{x \rightarrow 0} (1+2x)^{\frac{3}{x}} = e^6$$

$$44) \lim_{x \rightarrow 0} (1+2x)^{-\frac{2}{3x}} = e^{-\frac{4}{3}}$$

$$45) \lim_{x \rightarrow 0} (1-5x)^{-\frac{2}{3x}} = e^{\frac{10}{3}}$$

$$46) \boxed{\lim_{x \rightarrow \infty} \left(1 + \frac{1}{ax} \right)^{ax} = e} \text{ Limite Notable.}$$

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{2x}\right)^x = \lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^{\frac{t}{2}} = \left[\lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^t \right]^{\frac{1}{2}} = e^{\frac{1}{2}} = \sqrt{e}$$

$\begin{cases} \frac{1}{t} = \frac{1}{2x} \\ 2x = \frac{t}{2} \\ x = \frac{t}{2} \end{cases}$

$$47) \lim_{x \rightarrow \infty} \left(1 - \frac{1}{3x}\right)^x = e^{-\frac{1}{3}}$$

$$\frac{1}{t} = -\frac{1}{3x}$$

$$48) \lim_{x \rightarrow \infty} \left(1 + \frac{3}{7x}\right)^x = e^{\frac{3}{7}}$$

$$49) \lim_{x \rightarrow \infty} \left(1 - \frac{9}{x}\right)^{-3x} = e^{27}$$

$$50) \lim_{x \rightarrow \infty} \left(1 - \frac{3}{x}\right)^{4x} = e^{-12}$$

$$51) \lim_{x \rightarrow \infty} \left(1 + \frac{1}{5x}\right)^{-2x} = e^{-\frac{2}{5}}$$

$$52) \lim_{x \rightarrow \infty} \left(1 + \frac{1}{4x}\right)^{8x+1} = \text{Producto de potencias de igual base se suman los exponentes.}$$

$$\begin{aligned} \lim_{x \rightarrow \infty} \left(1 + \frac{1}{4x}\right)^{8x} \cdot \left(1 + \frac{1}{4x}\right) &= \lim_{x \rightarrow \infty} \left(1 + \frac{1}{4x}\right)^{8x} \cdot \lim_{x \rightarrow \infty} \left(1 + \frac{1}{4x}\right) = \\ &= \lim_{x \rightarrow \infty} \left(1 + \frac{1}{4x}\right)^{8x} \cdot 1 = \lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^{2t} = \left[\lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^t \right]^2 = \\ &= \left[\lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^t \right]^2 = e^2 \end{aligned}$$

$\begin{cases} \frac{1}{t} = \frac{1}{4x} \\ t = 4x \\ x = \frac{t}{4} \end{cases}$

$$53) \lim_{x \rightarrow \infty} \left(1 + \frac{4}{3x}\right)^{x-2} = \text{Cociente de potencias de igual base se restan los exponentes.}$$

$$\lim_{x \rightarrow \infty} \frac{\left(1 + \frac{4}{3x}\right)^x}{\left(1 + \frac{4}{3x}\right)^2} = \frac{\lim_{x \rightarrow \infty} \left(1 + \frac{4}{3x}\right)^x}{\lim_{x \rightarrow \infty} \left(1 + \frac{4}{3x}\right)^2} = \frac{\lim_{x \rightarrow \infty} \left(1 + \frac{4}{3x}\right)^x}{1}$$

$$= \lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^{\frac{4}{3}t} = \left[\lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^t \right]^{\frac{4}{3}} = e^{\frac{4}{3}}$$

$$\frac{1}{t} = \frac{4}{3x} \rightarrow x = \frac{4}{3}t$$

$$54) \lim_{x \rightarrow \infty} \left(1 + \frac{3}{2x}\right)^{x+1}$$

$$55) \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x-2}\right)^{4x} = e^4$$

$$\begin{cases} \frac{1}{t} = \frac{1}{x-2} \\ x-2 = t \\ x = t+2 \end{cases}$$

$$56) \lim_{x \rightarrow \infty} \left(1 + \frac{1}{1+x}\right)^{3x-2} = e^3$$

$$57) \lim_{x \rightarrow \infty} \left(1 + \frac{2}{x-1}\right)^{2x+3} = e^4$$

$$58) \lim_{x \rightarrow \infty} \left(1 - \frac{1}{x}\right)^{x^2} = \infty$$

$$59) \lim_{x \rightarrow \infty} \left(1 - \frac{3}{x}\right)^{x^2} = \infty$$

$$60) \lim_{x \rightarrow \infty} \left(1 - \frac{1}{x+2}\right)^{x^2+x} = \infty$$

$$\lim_{x \rightarrow \infty} \left(1 - \frac{1}{x+2}\right)^{x^2} \cdot \lim_{x \rightarrow \infty} \left(1 - \frac{1}{x+2}\right)^x$$

$$\frac{1}{t} = -\frac{1}{x+2} \rightarrow x = -t-2$$

$$= \lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^{(-t-2)^2} \cdot \lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^{-t-2} =$$

$$= \lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^{t^2 + 2t + 4} \cdot \lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^{-t-2} =$$

$$= \lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^{t^2} \cdot \lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^{2t} \cdot \lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^4 \cdot \frac{\lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^{-t}}{\lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^2} =$$

$$= \lim_{t \rightarrow \infty} \left[\left(1 + \frac{1}{t}\right)^t \right]^t \cdot \lim_{t \rightarrow \infty} \left[\left(1 + \frac{1}{t}\right)^t \right]^2 \cdot 1^4 \cdot \frac{\lim_{t \rightarrow \infty} \left[\left(1 + \frac{1}{t}\right)^t \right]^{-1}}{1} =$$

$$= \left[\lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^t \right]^{\lim_{t \rightarrow \infty} t} \cdot \left[\lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^t \right]^4 \cdot \frac{\left[\lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^t \right]^{-1}}{1} =$$

$$= e^\infty \cdot e^2 \cdot 1 \cdot \frac{e^{-1}}{1} = \infty_{vv}$$

$$61) \lim_{x \rightarrow \infty} \left(1 - \frac{3}{1+x}\right)^{2x^2-1} = \infty_{vv}$$

RESOLUCION DE LIMITES NOTABLES.

31) $\lim_{x \rightarrow 0} \frac{10x}{\text{sen } 3x} = \lim_{x \rightarrow 0} \frac{1}{\frac{\text{sen } 3x}{10x}} = \text{el límite de un cociente es igual al cociente de los límites}$

$$= \frac{\lim_{x \rightarrow 0} 1}{\lim_{x \rightarrow 0} \frac{\text{sen } 3x}{10x}} = \frac{1}{\lim_{x \rightarrow 0} \frac{\text{sen } 3x}{3 \cdot 10x}} = \frac{1}{\lim_{x \rightarrow 0} \left(\frac{3}{10} \cdot \frac{\text{sen } 3x}{3x} \right)} = \frac{1}{\frac{3}{10} \cdot \lim_{x \rightarrow 0} \frac{\text{sen } 3x}{3x}}$$

↑
LÍMITE NOTABLE

$$= \frac{1}{\frac{3}{10} \cdot 1} = \boxed{\frac{10}{3}} \checkmark$$

32) $\lim_{x \rightarrow 0} \frac{\text{sen } x}{5x} = \lim_{x \rightarrow 0} \left(\frac{1}{5} \cdot \frac{\text{sen } x}{x} \right) = \frac{1}{5} \cdot \lim_{x \rightarrow 0} \frac{\text{sen } x}{x} = \frac{1}{5} \cdot 1 = \boxed{\frac{1}{5}} \checkmark$

↓
LÍMITE NOTABLE

33) $\lim_{x \rightarrow 0} \frac{\text{sen } 2x}{x} = \lim_{x \rightarrow 0} \left(2 \cdot \frac{\text{sen } 2x}{2x} \right) = 2 \cdot \lim_{x \rightarrow 0} \frac{\text{sen } 2x}{2x} = 2 \cdot 1 = \boxed{2} \checkmark$

↓
LÍMITE NOTABLE

35) $\lim_{x \rightarrow 0} \frac{\text{sen}^2 5x}{15x^2} = \frac{1}{15} \lim_{x \rightarrow 0} \frac{\text{sen}^2 5x}{x^2} = \frac{1}{15} \lim_{x \rightarrow 0} \left(\frac{\text{sen } 5x}{x} \right)^2 =$

= EL LÍMITE DE UNA POTENCIA ES IGUAL A LA POTENCIA DEL LÍMITE =

$$= \frac{1}{15} \left(\lim_{x \rightarrow 0} \frac{\text{sen } 5x}{x} \right)^2 = \frac{1}{15} \left(\lim_{x \rightarrow 0} \frac{5}{5} \cdot \frac{\text{sen } 5x}{x} \right)^2 = \frac{1}{15} \left(5 \lim_{x \rightarrow 0} \frac{\text{sen } 5x}{5x} \right)^2 =$$

$$= \frac{1}{15} \cdot 5^2 \left(\lim_{x \rightarrow 0} \frac{\text{sen } 5x}{5x} \right)^2 = \frac{25}{15} \cdot \boxed{\frac{1}{3}} \checkmark$$

LÍMITE NOTABLE

$$43) \lim_{x \rightarrow 0} (1 + 2x)^{\frac{3}{x}} =$$

$$t = 2x$$

$$x = \frac{t}{2}$$

$$= \lim_{t \rightarrow 0} (1 + t)^{\frac{\frac{3}{t}}{2}} = \lim_{t \rightarrow 0} (1 + t)^{\frac{6}{t}} = \lim_{t \rightarrow 0} \left[(1 + t)^{\frac{1}{t}} \right]^6 =$$

$$= \left[\lim_{t \rightarrow 0} (1 + t)^{\frac{1}{t}} \right]^6 = \boxed{e^6} \checkmark \checkmark$$

$$44) \lim_{x \rightarrow 0} (1 + 2x)^{-\frac{2}{3x}} =$$

$$t = 2x$$

$$x = \frac{t}{2}$$

$$= \lim_{t \rightarrow 0} (1 + t)^{-\frac{2}{3(\frac{t}{2})}} = \lim_{t \rightarrow 0} (1 + t)^{-\frac{4}{3t}} = \lim_{t \rightarrow 0} \left[(1 + t)^{\frac{1}{t}} \right]^{-\frac{4}{3}} =$$

$$= \left[\lim_{t \rightarrow 0} (1 + t)^{\frac{1}{t}} \right]^{-\frac{4}{3}} = e^{-\frac{4}{3}} = \frac{1}{e^{\frac{4}{3}}} = \frac{1}{\sqrt[3]{e^4}} = \text{Racionalizamos:}$$

$$= \frac{1 \cdot \sqrt[3]{e^2}}{\sqrt[3]{e^4} \cdot \sqrt[3]{e^2}} = \frac{\sqrt[3]{e^2}}{\sqrt[3]{e^4 \cdot e^2}} = \frac{\sqrt[3]{e^2}}{\sqrt[3]{e^6}} = \frac{\sqrt[3]{e^2}}{e^2}$$

$$45) \lim_{x \rightarrow 0} (1 - 5x)^{-\frac{2}{3x}} =$$

$$t = -5x$$

$$x = -\frac{t}{5}$$

$$= \lim_{t \rightarrow 0} (1 + t)^{-\frac{2}{3(-\frac{t}{5})}} = \lim_{t \rightarrow 0} (1 + t)^{\frac{10}{3t}} = \lim_{t \rightarrow 0} \left[(1 + t)^{\frac{1}{t}} \right]^{\frac{10}{3}} =$$

$$= \left[\lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^{\frac{1}{t}} \right]^{\frac{10}{3}} = e^{\frac{10}{3}} = \sqrt[3]{e^{10}} \quad \text{V.V.}$$

47) $\lim_{x \rightarrow \infty} \left(1 - \frac{1}{3x}\right)^x =$ EN ESTE CASO EL REEMPLAZO O CAMBIO DE VARIABLE SERA $\frac{1}{t}$:

$$\frac{1}{t} = -\frac{1}{3x}$$

$$x = -\frac{t}{3}$$

$$= \lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^{-\frac{t}{3}} = \lim_{t \rightarrow \infty} \left[\left(1 + \frac{1}{t}\right)^t \right]^{-\frac{1}{3}} = \left[\lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^t \right]^{-\frac{1}{3}} =$$

$$= e^{-\frac{1}{3}} = \frac{1}{e^{\frac{1}{3}}} = \frac{1}{\sqrt[3]{e}} = \frac{1 \cdot \sqrt[3]{e^2}}{\sqrt[3]{e} \cdot \sqrt[3]{e^2}} = \frac{\sqrt[3]{e^2}}{\sqrt[3]{e^3}} = \frac{\sqrt[3]{e^2}}{e} \quad \text{V.V.}$$

48) $\lim_{t \rightarrow \infty} \left(1 + \frac{3}{7t}\right)^x =$

$$\frac{1}{t} = \frac{3}{7x}$$

$$x = \frac{3t}{7}$$

$$\lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^{\frac{3}{7}t} = \lim_{t \rightarrow \infty} \left[\left(1 + \frac{1}{t}\right)^t \right]^{\frac{3}{7}} = \left[\lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^t \right]^{\frac{3}{7}} = e^{\frac{3}{7}} = \sqrt[7]{e^3} \quad \text{V.V.}$$

49) $\lim_{t \rightarrow \infty} \left(1 - \frac{9}{x}\right)^{-3x} =$

$$\frac{1}{t} = -\frac{9}{x}$$

$$x = -9t$$

$$\lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^{(-3)(-9t)} = \lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^{27t} = \lim_{t \rightarrow \infty} \left[\left(1 + \frac{1}{t}\right)^t \right]^{27} = \left[\lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^t \right]^{27} =$$

$$= e^{27} \quad \text{V.V.}$$

$$66) \lim_{x \rightarrow \infty} \left(\frac{3x}{3x-4} \right)^{\frac{x^2+2}{x-1}} =$$

$$\left(\frac{x^2+2}{x-1} = x+1 + \frac{3}{x-1} \right)$$

$$= \lim_{x \rightarrow \infty} \left(\frac{\frac{3x}{3x}}{\frac{3x-4}{3x}} \right)^{x+1 + \frac{3}{x-1}} = \lim_{x \rightarrow \infty} \left(\frac{1}{1 - \frac{4}{3x}} \right)^{x+1 + \frac{3}{x-1}}$$

$$= \lim_{x \rightarrow \infty} \left[\left(\frac{1}{1 - \frac{4}{3x}} \right)^{x+1} \cdot \left(\frac{1}{1 - \frac{4}{3x}} \right)^{\frac{3}{x-1}} \right] =$$

$$= \lim_{x \rightarrow \infty} \left(\frac{1}{1 - \frac{4}{3x}} \right)^{x+1} \cdot \lim_{x \rightarrow \infty} \left(\frac{1}{1 - \frac{4}{3x}} \right)^{\frac{3}{x-1}} =$$

$$= \lim_{x \rightarrow \infty} \left[\left(1 - \frac{4}{3x} \right)^{-1} \right]^{x+1} \cdot \lim_{x \rightarrow \infty} \left[\left(1 - \frac{4}{3x} \right)^{-1} \right]^{\frac{3}{x-1}}$$

$$-\frac{4}{3x} = \frac{1}{t} \therefore t = -\frac{3x}{4} \rightarrow x = -\frac{4}{3}t$$

$$\therefore x+1 = -\frac{4}{3}t + 1$$

$$\frac{3}{x-1} = \frac{3}{-\frac{4}{3}t - 1}$$

$$= \lim_{t \rightarrow \infty} \left[\left(1 + \frac{1}{t} \right)^{-1} \right]^{-\frac{4}{3}t+1} \cdot \lim_{t \rightarrow \infty} \left[\left(1 + \frac{1}{t} \right)^{-1} \right]^{\frac{3}{-\frac{4}{3}t-1}} =$$

$$= \lim_{t \rightarrow \infty} \left[\left(1 + \frac{1}{t} \right)^{(-1)(-\frac{4}{3}t)} \cdot \left(1 + \frac{1}{t} \right)^{(-1) \cdot 1} \right] \lim_{t \rightarrow \infty} \left[\left(1 + \frac{1}{t} \right)^{-1} \right]^{\lim_{t \rightarrow \infty} \frac{3}{-\frac{4}{3}t-1}} =$$

$$\begin{aligned}
 &= \lim_{t \rightarrow \infty} \left[\left(1 + \frac{1}{t}\right)^t \right]^{(-1) \cdot \left(-\frac{4}{3}\right)} \cdot \lim_{t \rightarrow \infty} \left[\left(1 + \frac{1}{t}\right) \right]^{(-1)} \cdot \left[\lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^{(-1)} \right]^{\lim_{t \rightarrow \infty} \frac{3}{-\frac{4}{3}t - 1}} \\
 &= \left[\lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^t \right]^{\frac{4}{3}} \cdot \left[\lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^{(-1)} \cdot (1) \right]^{\lim_{t \rightarrow \infty} \frac{3}{-\frac{4}{3}t - 1}} \\
 &= e^{\frac{4}{3}} \cdot 1^{(-1)} \cdot (1)^{\frac{3}{(-\infty)}} = e^{\frac{4}{3}} \cdot 1 \cdot 1 = e^{\frac{4}{3}}
 \end{aligned}$$

$$\begin{aligned}
 67) \quad \lim_{x \rightarrow \infty} \left(\frac{7x}{7x+2} \right)^{x-1} &= \lim_{x \rightarrow \infty} \left(\frac{1}{\frac{7x+2}{7x}} \right)^{x-1} = \lim_{x \rightarrow \infty} \frac{\left(\frac{1}{\frac{7x+2}{7x}} \right)^x}{\left(\frac{1}{\frac{7x+2}{7x}} \right)^1} \\
 &= \frac{\lim_{x \rightarrow \infty} \left(\frac{1}{\frac{7x+2}{7x}} \right)^x}{\lim_{x \rightarrow \infty} \left(\frac{1}{\frac{7x+2}{7x}} \right)} = \frac{\lim_{x \rightarrow \infty} \left(\frac{1}{1 + \frac{2}{7x}} \right)}{\lim_{x \rightarrow \infty} \left(\frac{1}{1 + \frac{2}{7x}} \right)} \\
 &= \frac{\lim_{x \rightarrow \infty} \left(\frac{1}{1 + \frac{2}{7x}} \right)^x}{1} = \begin{cases} \frac{2}{7x} = \frac{1}{t} \\ t = \frac{7}{2}x \\ x = \frac{2}{7}t \end{cases} \\
 &= \lim_{t \rightarrow \infty} \left[\left(\frac{1}{1 + \frac{1}{t}} \right)^{\frac{2}{7}t} \right] = \lim_{t \rightarrow \infty} \left[\left(1 + \frac{1}{t} \right)^{-1} \right]^{\frac{2}{7}t} \\
 &= \lim_{t \rightarrow \infty} \left[\left(1 + \frac{1}{t} \right)^t \right]^{(-1) \cdot \frac{2}{7}} = \lim_{t \rightarrow \infty} \left[\left(1 + \frac{1}{t} \right)^t \right]^{-\frac{2}{7}} = e^{-\frac{2}{7}}
 \end{aligned}$$

$$68) \lim_{x \rightarrow 0} \frac{x}{1 - \cos x} = \lim_{x \rightarrow 0} \frac{x(1 + \cos x)}{(1 - \cos x)(1 + \cos x)} = \lim_{x \rightarrow 0} \frac{x(1 + \cos x)}{1 - \cos^2 x}$$

$$= \lim_{x \rightarrow 0} \frac{x + x \cos x}{\sin^2 x} = \begin{cases} \sin^2 x + \cos^2 x = 1 \\ \sin^2 x = 1 - \cos^2 x \end{cases}$$

$$= \lim_{x \rightarrow 0} \left(\frac{x}{\sin^2 x} + \frac{x \cos x}{\sin^2 x} \right) = \lim_{x \rightarrow 0} \frac{x}{\sin x \cdot \sin x} + \lim_{x \rightarrow 0} \frac{x \cdot \cos x}{\sin x \cdot \sin x}$$

$$= \lim_{x \rightarrow 0} \left(\frac{x}{\sin x} \cdot \frac{1}{\sin x} \right) + \lim_{x \rightarrow 0} \left(\frac{x}{\sin x} \cdot \frac{\cos x}{\sin x} \right) =$$

$$= \left(\lim_{x \rightarrow 0} \frac{x}{\sin x} \cdot \lim_{x \rightarrow 0} \frac{1}{\sin x} \right) + \left(\lim_{x \rightarrow 0} \frac{x}{\sin x} \cdot \lim_{x \rightarrow 0} \frac{\cos x}{\sin x} \right) =$$

$$= \left(1 \cdot \frac{1}{0} \right) + \left(1 \cdot \lim_{x \rightarrow 0} \frac{\frac{1}{\sin x}}{\frac{\cos x}{\sin x}} \right) = \infty + \lim_{x \rightarrow 0} \frac{1}{\tan x} =$$

$$= \infty + \infty = \infty$$